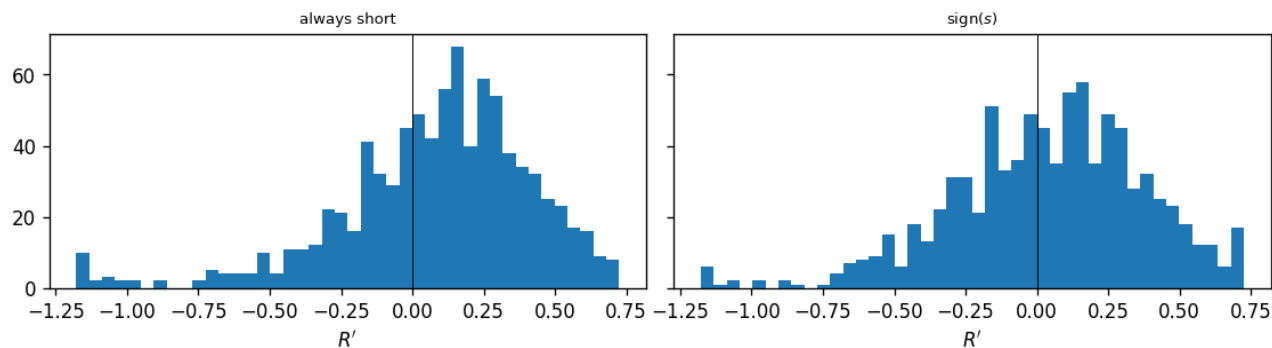


0DTE nearest-OTM straddle: rule table by strategy, every intraday entry window — delta-hedged, held to cash-settle

Rule table by strategy — entry 10:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{clk}	n_{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	864	0.078	0.371	-2.95	0.89	-1.96	9.38	6.15	3.32	0	0.0	2.29
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	864	0.035	0.378	-2.95	1.48	-1.12	7.64	2.75	1.48	307	35.5	0.48
block-diagonal ridge	864	0.039	0.377	-2.95	1.48	-1.13	7.72	3.06	1.66	303	35.1	0.65
block-diagonal ridge, without the FOMC columns	864	0.035	0.378	-2.95	1.48	-1.10	7.62	2.70	1.46	312	36.1	0.45
LightGBM [†]	864	0.048	0.376	-2.95	1.48	-1.16	7.93	3.78	2.04	286	33.1	1.03
XGBoost [†]	864	0.045	0.377	-2.95	1.48	-1.16	7.84	3.48	1.88	281	32.5	0.87
lasso (causally tuned) [†]	864	0.031	0.378	-2.95	1.48	-0.96	7.51	2.44	1.32	310	35.9	0.31
lasso ($\alpha = 10^{-4}$)	864	0.038	0.377	-2.95	1.48	-1.11	7.68	2.95	1.59	294	34.0	0.59
elastic net (causally tuned) [†]	864	0.032	0.378	-2.95	1.48	-1.08	7.57	2.52	1.36	314	36.3	0.36

block-diagonal ridge, midpoint fills: entry 10:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



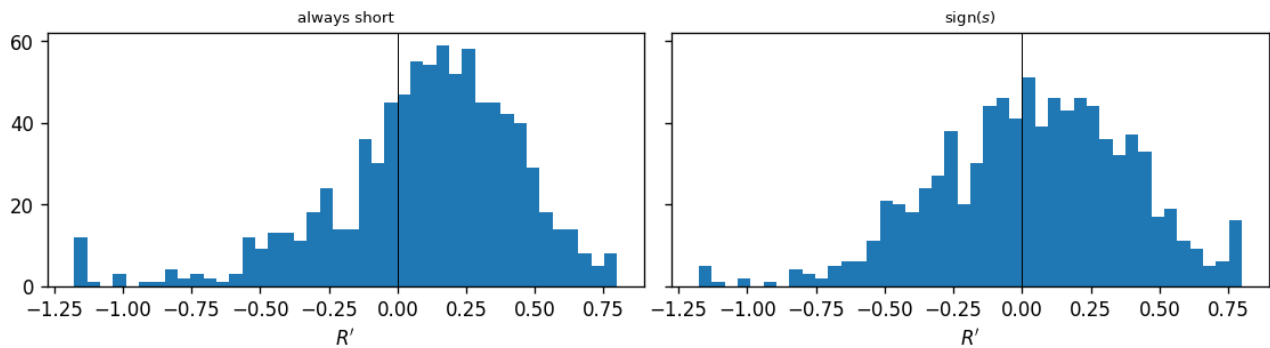
Distribution of the return of the position taken at 10:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 864 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 10:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.096	0.366	-2.00	0.88	-1.44	4.42	7.74	4.18	0	0.0	3.51
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.036	0.377	-1.81	2.00	-0.22	2.66	2.81	1.52	289	33.4	0.87
block-diagonal ridge	865	0.046	0.376	-1.81	2.00	-0.02	2.61	3.56	1.92	301	34.8	1.27
block-diagonal ridge, without the FOMC columns	865	0.049	0.376	-1.81	2.00	-0.01	2.62	3.82	2.06	311	36.0	1.41
LightGBM [†]	865	0.043	0.376	-1.81	2.00	-0.07	2.63	3.35	1.81	295	34.1	1.16
XGBoost [†]	865	0.049	0.376	-1.81	2.00	-0.09	2.66	3.80	2.05	287	33.2	1.40
lasso (causally tuned) [†]	865	0.044	0.376	-1.81	2.00	-0.12	2.66	3.43	1.85	310	35.8	1.20
lasso ($\alpha = 10^{-4}$)	865	0.051	0.375	-1.81	2.00	-0.05	2.65	4.01	2.16	287	33.2	1.51
elastic net (causally tuned) [†]	865	0.046	0.376	-1.81	2.00	-0.13	2.67	3.62	1.95	297	34.3	1.30

block-diagonal ridge, midpoint fills: entry 10:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



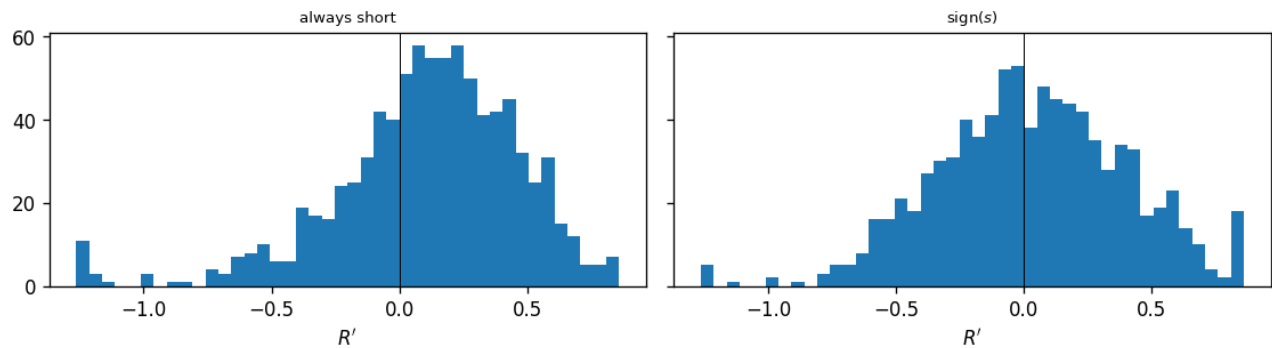
Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 11:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.103	0.388	-2.30	0.87	-1.57	5.53	7.78	4.20	0	0.0	3.58
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.044	0.399	-2.09	2.30	0.19	3.36	3.25	1.75	364	42.1	1.15
block-diagonal ridge	865	0.033	0.400	-2.09	2.30	0.14	3.36	2.46	1.33	365	42.2	0.73
block-diagonal ridge, without the FOMC columns	865	0.029	0.401	-2.09	2.30	0.14	3.35	2.10	1.13	378	43.7	0.53
LightGBM [†]	865	0.030	0.401	-2.09	2.30	0.18	3.34	2.21	1.19	362	41.8	0.59
XGBoost [†]	865	0.038	0.400	-2.09	2.30	0.19	3.35	2.80	1.51	370	42.8	0.91
lasso (causally tuned) [†]	865	0.033	0.400	-2.09	2.30	0.14	3.36	2.45	1.32	379	43.8	0.72
lasso ($\alpha = 10^{-4}$)	865	0.034	0.400	-2.09	2.30	0.13	3.36	2.47	1.33	362	41.8	0.73
elastic net (causally tuned) [†]	865	0.033	0.400	-2.09	2.30	0.12	3.36	2.45	1.32	374	43.2	0.72

block-diagonal ridge, midpoint fills: entry 11:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



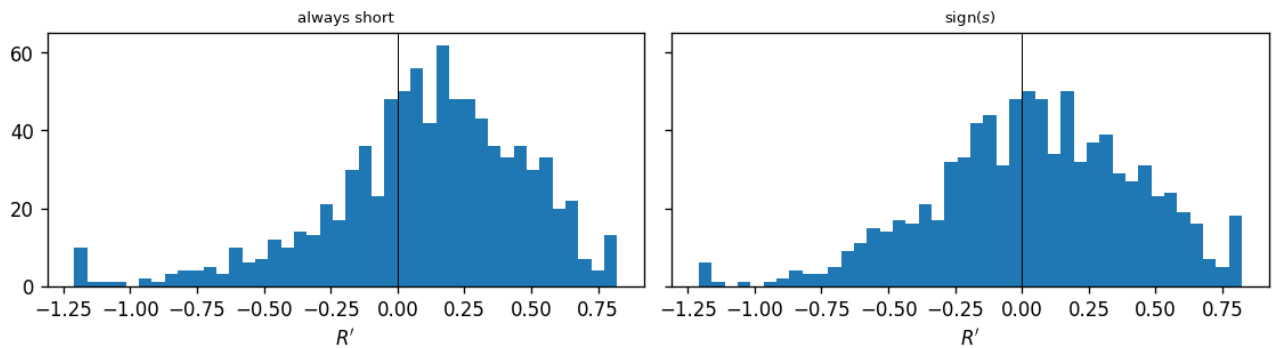
Distribution of the return of the position taken at 11:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 11:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.101	0.407	-3.05	0.89	-1.81	8.19	7.32	3.95	0	0.0	3.33
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.046	0.416	-2.31	3.05	0.05	5.77	3.26	1.76	333	38.5	1.15
block-diagonal ridge	865	0.052	0.416	-2.31	3.05	0.04	5.81	3.65	1.97	336	38.8	1.36
block-diagonal ridge, without the FOMC columns	865	0.050	0.416	-2.31	3.05	0.24	5.70	3.56	1.92	349	40.3	1.31
LightGBM [†]	865	0.039	0.417	-2.31	3.05	0.06	5.73	2.74	1.48	333	38.5	0.87
XGBoost [†]	865	0.042	0.417	-2.31	3.05	0.04	5.75	2.95	1.59	334	38.6	0.99
lasso (causally tuned) [†]	865	0.041	0.417	-2.31	3.05	0.08	5.73	2.92	1.58	344	39.8	0.97
lasso ($\alpha = 10^{-4}$)	865	0.041	0.417	-2.31	3.05	0.07	5.73	2.87	1.55	351	40.6	0.94
elastic net (causally tuned) [†]	865	0.047	0.416	-2.31	3.05	0.07	5.77	3.31	1.78	347	40.1	1.18

block-diagonal ridge, midpoint fills: entry 11:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



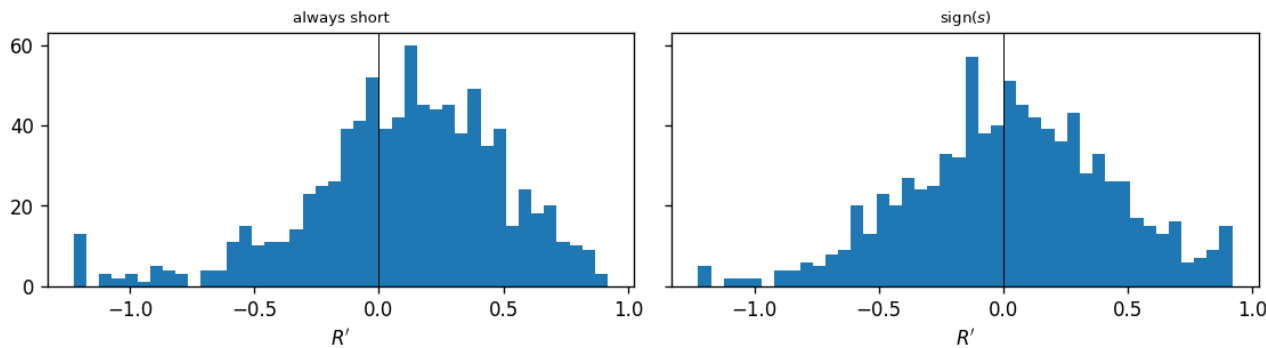
Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 12:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.087	0.434	-2.97	0.93	-1.48	5.45	5.88	3.17	0	0.0	2.58
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.025	0.442	-2.97	1.98	-0.67	4.12	1.68	0.91	364	42.0	0.33
block-diagonal ridge	866	0.033	0.442	-2.06	2.97	0.26	3.91	2.22	1.20	367	42.4	0.62
block-diagonal ridge, without the FOMC columns	866	0.030	0.442	-2.06	2.97	0.27	3.90	2.03	1.09	379	43.8	0.51
LightGBM [†]	866	0.040	0.441	-2.10	2.97	0.09	3.97	2.67	1.44	360	41.6	0.86
XGBoost [†]	866	0.023	0.442	-2.06	2.97	0.07	3.95	1.52	0.82	372	43.0	0.24
lasso (causally tuned) [†]	866	0.025	0.442	-2.97	1.98	-0.60	4.10	1.64	0.88	385	44.5	0.30
lasso ($\alpha = 10^{-4}$)	866	0.036	0.441	-2.06	2.97	0.22	3.92	2.39	1.29	372	43.0	0.71
elastic net (causally tuned) [†]	866	0.025	0.442	-2.10	2.97	0.08	3.95	1.67	0.90	382	44.1	0.32

block-diagonal ridge, midpoint fills: entry 12:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



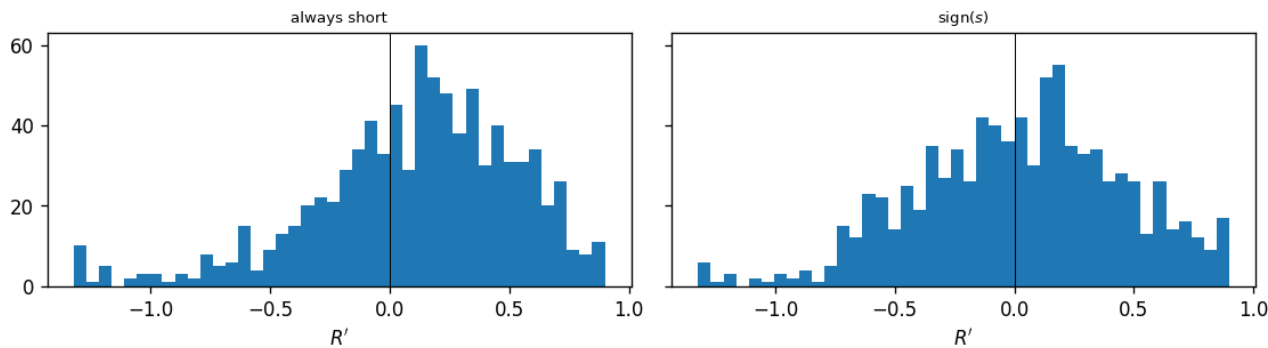
Distribution of the return of the position taken at 12:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 12:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.097	0.466	-4.31	0.93	-1.88	10.76	6.12	3.30	0	0.0	2.72
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.021	0.475	-2.37	4.31	0.59	8.36	1.31	0.71	379	43.8	0.14
block-diagonal ridge	866	0.027	0.475	-2.37	4.31	0.53	8.36	1.69	0.91	369	42.6	0.34
block-diagonal ridge, without the FOMC columns	866	0.024	0.475	-2.37	4.31	0.61	8.35	1.47	0.79	395	45.6	0.22
LightGBM [†]	866	0.024	0.475	-2.37	4.31	0.63	8.34	1.48	0.80	385	44.5	0.23
XGBoost [†]	866	0.024	0.475	-2.37	4.31	0.61	8.35	1.47	0.80	391	45.2	0.22
lasso (causally tuned) [†]	866	0.023	0.475	-2.37	4.31	0.56	8.36	1.44	0.78	406	46.9	0.20
lasso ($\alpha = 10^{-4}$)	866	0.024	0.475	-2.37	4.31	0.53	8.36	1.50	0.81	371	42.8	0.24
elastic net (causally tuned) [†]	866	0.027	0.475	-2.37	4.31	0.54	8.36	1.67	0.90	389	44.9	0.33

block-diagonal ridge, midpoint fills: entry 12:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



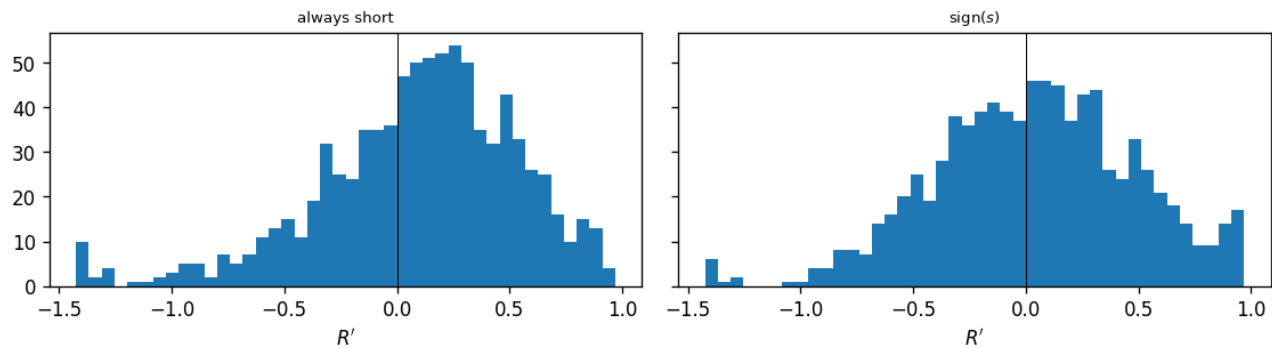
Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 13:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.083	0.498	-5.47	0.95	-2.48	19.28	4.93	2.66	0	0.0	2.09
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.026	0.504	-2.44	5.47	1.38	16.47	1.51	0.82	446	51.5	0.26
block-diagonal ridge	866	0.044	0.503	-2.44	5.47	1.15	16.52	2.55	1.37	418	48.3	0.81
block-diagonal ridge, without the FOMC columns	866	0.037	0.503	-2.44	5.47	1.18	16.50	2.17	1.17	435	50.2	0.61
LightGBM [†]	866	0.027	0.504	-2.44	5.47	1.36	16.47	1.59	0.86	447	51.6	0.30
XGBoost [†]	866	0.029	0.504	-2.44	5.47	1.35	16.47	1.68	0.91	452	52.2	0.35
lasso (causally tuned) [†]	866	0.029	0.504	-2.44	5.47	1.19	16.50	1.69	0.91	435	50.2	0.35
lasso ($\alpha = 10^{-4}$)	866	0.039	0.503	-2.44	5.47	1.17	16.50	2.27	1.23	421	48.6	0.66
elastic net (causally tuned) [†]	866	0.025	0.504	-2.44	5.47	1.20	16.51	1.49	0.80	438	50.6	0.24

block-diagonal ridge, midpoint fills: entry 13:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



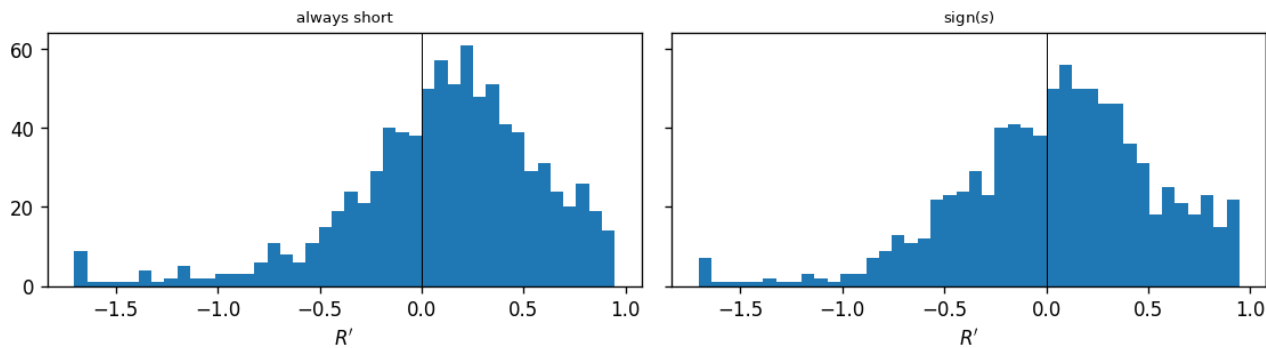
Distribution of the return of the position taken at 13:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 13:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.088	0.557	-6.72	1.00	-3.24	28.71	4.65	2.51	0	0.0	1.97
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.057	0.561	-2.77	6.72	1.78	25.16	2.99	1.61	306	35.3	1.08
block-diagonal ridge	866	0.058	0.561	-3.14	6.72	1.32	25.36	3.02	1.63	270	31.2	1.09
block-diagonal ridge, without the FOMC columns	866	0.052	0.561	-3.14	6.72	1.35	25.30	2.72	1.47	292	33.7	0.93
LightGBM [†]	866	0.043	0.562	-3.14	6.72	1.43	25.22	2.27	1.22	300	34.6	0.69
XGBoost [†]	866	0.056	0.561	-3.14	6.72	1.43	25.30	2.93	1.58	303	35.0	1.05
lasso (causally tuned) [†]	866	0.048	0.562	-3.14	6.72	1.40	25.25	2.50	1.35	302	34.9	0.81
lasso ($\alpha = 10^{-4}$)	866	0.055	0.561	-3.14	6.72	1.33	25.33	2.89	1.56	267	30.8	1.02
elastic net (causally tuned) [†]	866	0.050	0.562	-3.14	6.72	1.38	25.27	2.62	1.42	295	34.1	0.88

block-diagonal ridge, midpoint fills: entry 13:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



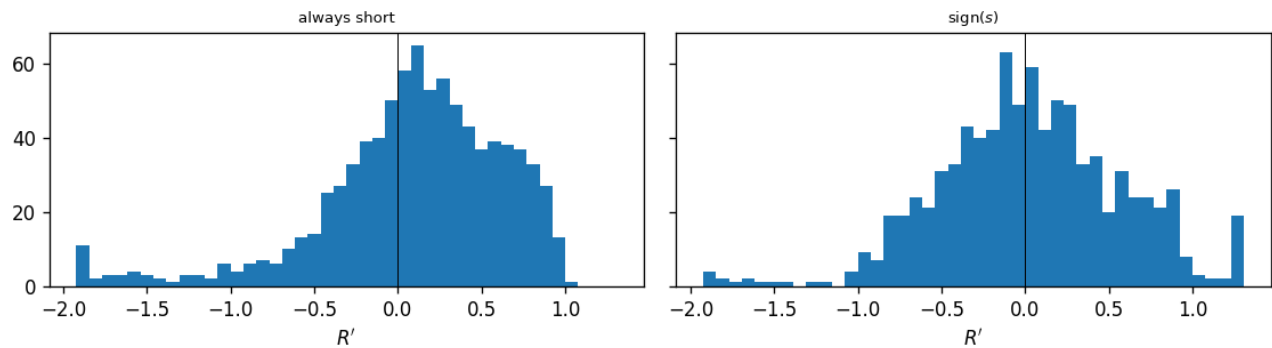
Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 14:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.080	0.623	-6.94	1.00	-2.74	20.45	3.77	2.04	0	0.0	1.36
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.026	0.628	-2.93	6.94	1.45	18.25	1.22	0.66	451	52.1	-0.01
block-diagonal ridge	866	0.041	0.627	-2.93	6.94	1.41	18.22	1.93	1.04	427	49.3	0.38
block-diagonal ridge, without the FOMC columns	866	0.034	0.627	-2.76	6.94	1.64	18.18	1.59	0.86	442	51.0	0.19
LightGBM [†]	866	0.039	0.627	-2.93	6.94	1.61	18.17	1.83	0.99	469	54.2	0.33
XGBoost [†]	866	0.032	0.627	-2.93	6.94	1.63	18.19	1.51	0.82	472	54.5	0.15
lasso (causally tuned) [†]	866	0.022	0.628	-2.93	6.94	1.39	18.28	1.02	0.55	450	52.0	-0.11
lasso ($\alpha = 10^{-4}$)	866	0.046	0.627	-2.93	6.94	1.41	18.21	2.16	1.16	424	49.0	0.50
elastic net (causally tuned) [†]	866	0.020	0.628	-2.93	6.94	1.42	18.29	0.93	0.50	446	51.5	-0.16

block-diagonal ridge, midpoint fills: entry 14:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



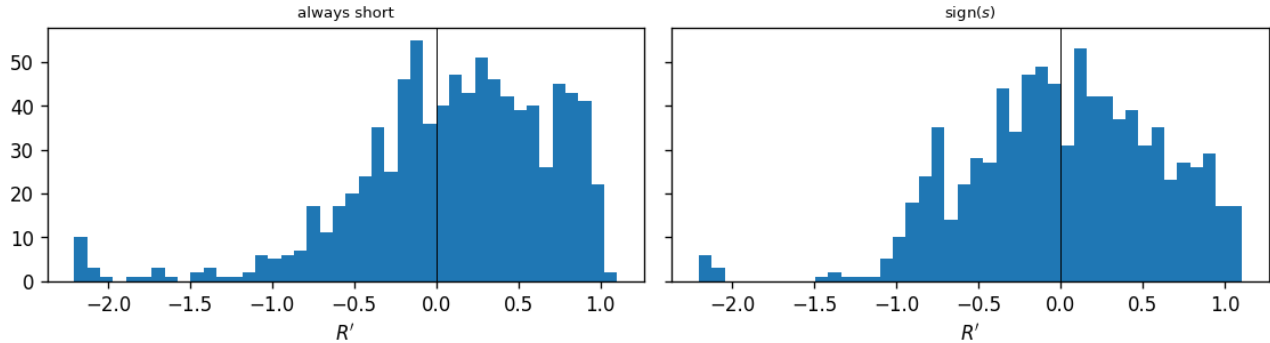
Distribution of the return of the position taken at 14:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 14:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.091	0.718	-8.43	1.05	-3.71	32.65	3.71	2.00	0	0.0	1.50
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.035	0.723	-6.46	8.43	1.07	29.74	1.41	0.76	335	38.7	0.26
block-diagonal ridge	866	0.031	0.723	-6.46	8.43	0.66	29.81	1.25	0.67	342	39.5	0.17
block-diagonal ridge, without the FOMC columns	866	0.033	0.723	-6.46	8.43	0.66	29.82	1.36	0.73	350	40.4	0.23
LightGBM [†]	866	0.045	0.722	-6.46	8.43	1.06	29.78	1.82	0.98	350	40.4	0.48
XGBoost [†]	866	0.043	0.722	-6.46	8.43	1.12	29.75	1.74	0.94	367	42.4	0.44
lasso (causally tuned) [†]	866	0.031	0.723	-6.46	8.43	1.15	29.72	1.27	0.68	361	41.7	0.18
lasso ($\alpha = 10^{-4}$)	866	0.021	0.723	-8.43	2.98	-2.97	30.22	0.87	0.47	337	38.9	-0.03
elastic net (causally tuned) [†]	866	0.042	0.722	-6.46	8.43	1.13	29.75	1.69	0.91	358	41.3	0.41

block-diagonal ridge, midpoint fills: entry 14:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



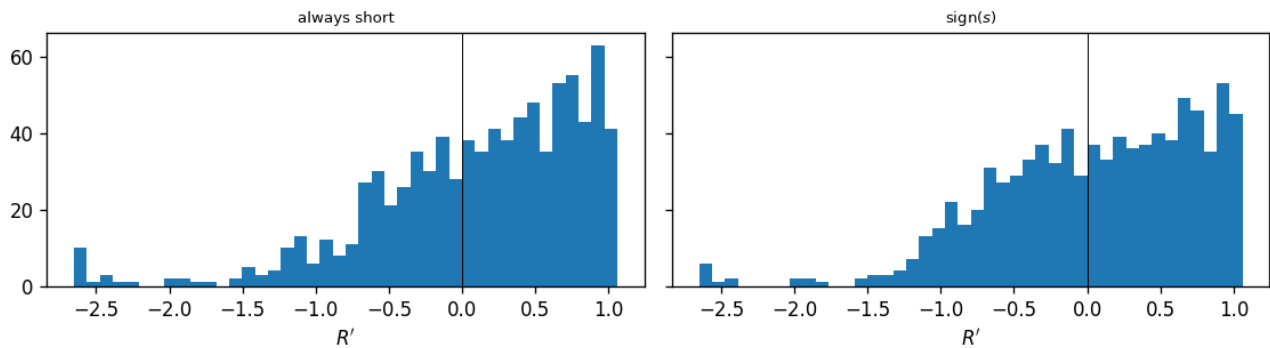
Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 15:00 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.095	0.778	-5.81	1.04	-1.95	7.83	3.59	1.93	0	0.0	1.40
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.030	0.783	-5.81	3.95	-1.25	6.88	1.12	0.60	201	23.2	0.07
block-diagonal ridge	866	0.065	0.781	-5.81	3.95	-0.97	7.10	2.47	1.33	186	21.5	0.80
block-diagonal ridge, without the FOMC columns	866	0.068	0.781	-5.81	3.95	-0.97	7.11	2.55	1.37	189	21.8	0.84
LightGBM [†]	866	0.053	0.782	-5.81	3.95	-0.94	7.00	2.01	1.08	205	23.7	0.55
XGBoost [†]	866	0.043	0.783	-5.81	3.95	-0.94	6.92	1.60	0.86	220	25.4	0.33
lasso (causally tuned) [†]	866	0.064	0.781	-5.81	3.95	-1.01	7.09	2.40	1.29	198	22.9	0.76
lasso ($\alpha = 10^{-4}$)	866	0.071	0.781	-5.81	3.95	-0.99	7.15	2.67	1.44	185	21.4	0.90
elastic net (causally tuned) [†]	866	0.047	0.782	-5.81	3.95	-1.12	7.00	1.78	0.96	190	21.9	0.43

block-diagonal ridge, midpoint fills: entry 15:00 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



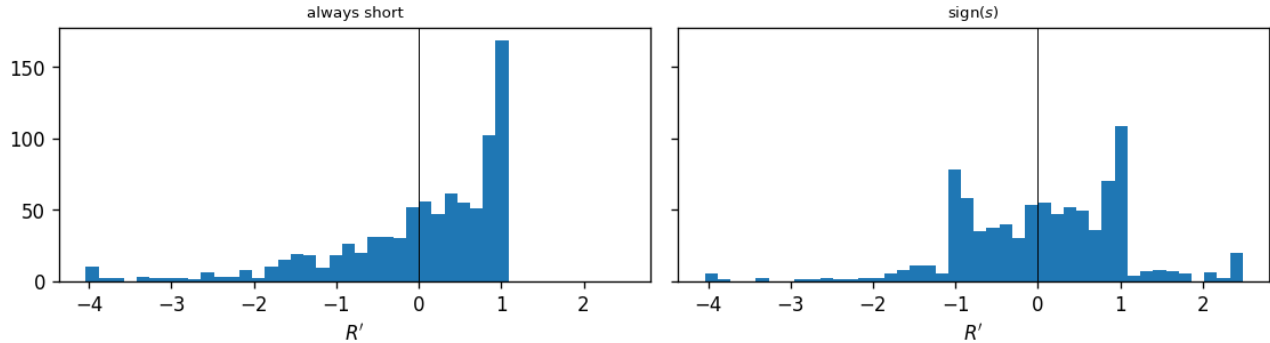
Distribution of the return of the position taken at 15:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 15:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 15:30 ET, held to cash-settle

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.017	1.115	-9.38	1.08	-2.26	9.18	0.44	0.24	0	0.0	-0.23
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.064	1.114	-6.37	9.38	0.12	9.07	1.70	0.92	286	33.0	0.44
block-diagonal ridge	866	0.091	1.112	-6.37	9.38	0.62	8.95	2.41	1.30	346	40.0	0.83
block-diagonal ridge, without the FOMC columns	866	0.092	1.112	-4.45	9.38	1.01	8.82	2.45	1.32	355	41.0	0.85
LightGBM [†]	866	0.104	1.111	-6.37	9.38	0.62	8.96	2.76	1.49	290	33.5	1.01
XGBoost [†]	866	0.102	1.111	-6.37	9.38	0.52	9.00	2.72	1.46	279	32.2	0.99
lasso (causally tuned) [†]	866	0.103	1.111	-6.37	9.38	0.40	9.04	2.74	1.48	321	37.1	1.00
lasso ($\alpha = 10^{-4}$)	866	0.105	1.111	-6.37	9.38	0.95	8.84	2.79	1.51	358	41.3	1.03
elastic net (causally tuned) [†]	866	0.075	1.113	-6.37	9.38	0.27	9.05	1.98	1.07	307	35.5	0.60

block-diagonal ridge, midpoint fills: entry 15:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



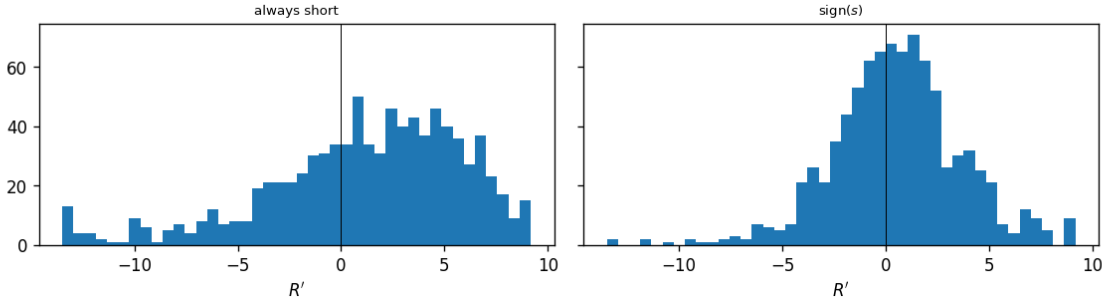
Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the package expires worthless).

One trade a day: enter the nearest-OTM straddle at 15:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

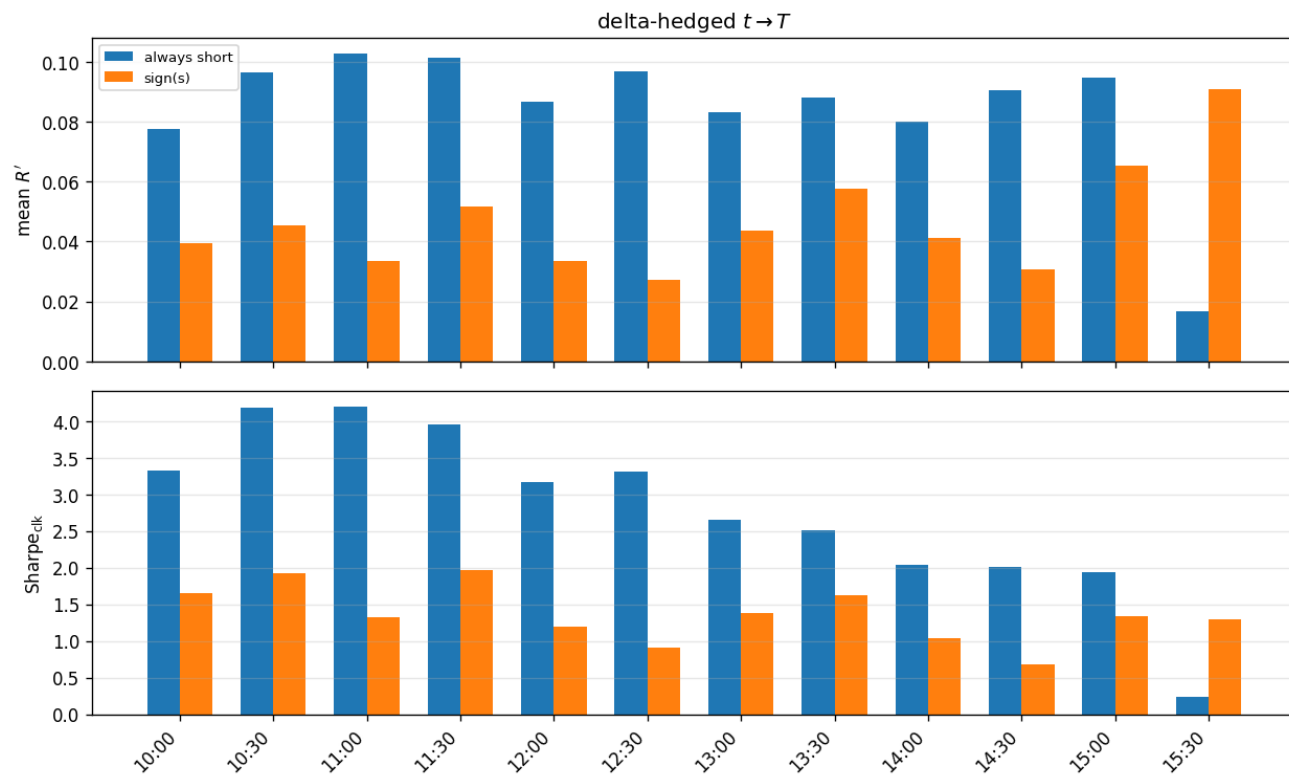
Rule table by strategy — pooled over the twelve entry windows

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	<i>n</i> _{buy}	%buy	Sharpe ^x _{ann}
<i>Short volatility (always short)</i>												
all models (no forecast)	866	1.014	5.487	-54.71	10.01	-2.40	14.95	5.44	2.93	0	0.0	2.21
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.446	3.633	-16.36	48.77	2.45	36.72	3.61	1.95	266	30.7	0.86
block-diagonal ridge	866	0.560	3.727	-20.53	54.71	3.21	52.60	4.42	2.38	252	29.1	1.33
block-diagonal ridge, without the FOMC columns	866	0.532	3.690	-16.36	54.71	3.55	53.83	4.25	2.29	267	30.8	1.22
LightGBM [†]	866	0.536	3.725	-20.53	54.71	3.32	52.85	4.23	2.28	277	32.0	1.23
XGBoost [†]	866	0.524	3.738	-20.53	54.71	3.30	52.17	4.13	2.23	271	31.3	1.17
lasso (causally tuned) [†]	866	0.494	3.658	-20.53	48.77	2.36	36.43	3.98	2.14	276	31.9	1.07
lasso ($\alpha = 10^{-4}$)	866	0.561	3.491	-20.53	37.84	1.00	16.98	4.73	2.55	248	28.6	1.42
elastic net (causally tuned) [†]	866	0.470	3.748	-20.53	54.71	3.12	51.80	3.69	1.99	266	30.7	0.94

block-diagonal ridge, midpoint fills: daily sum over the twelve entry windows 10:00-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)

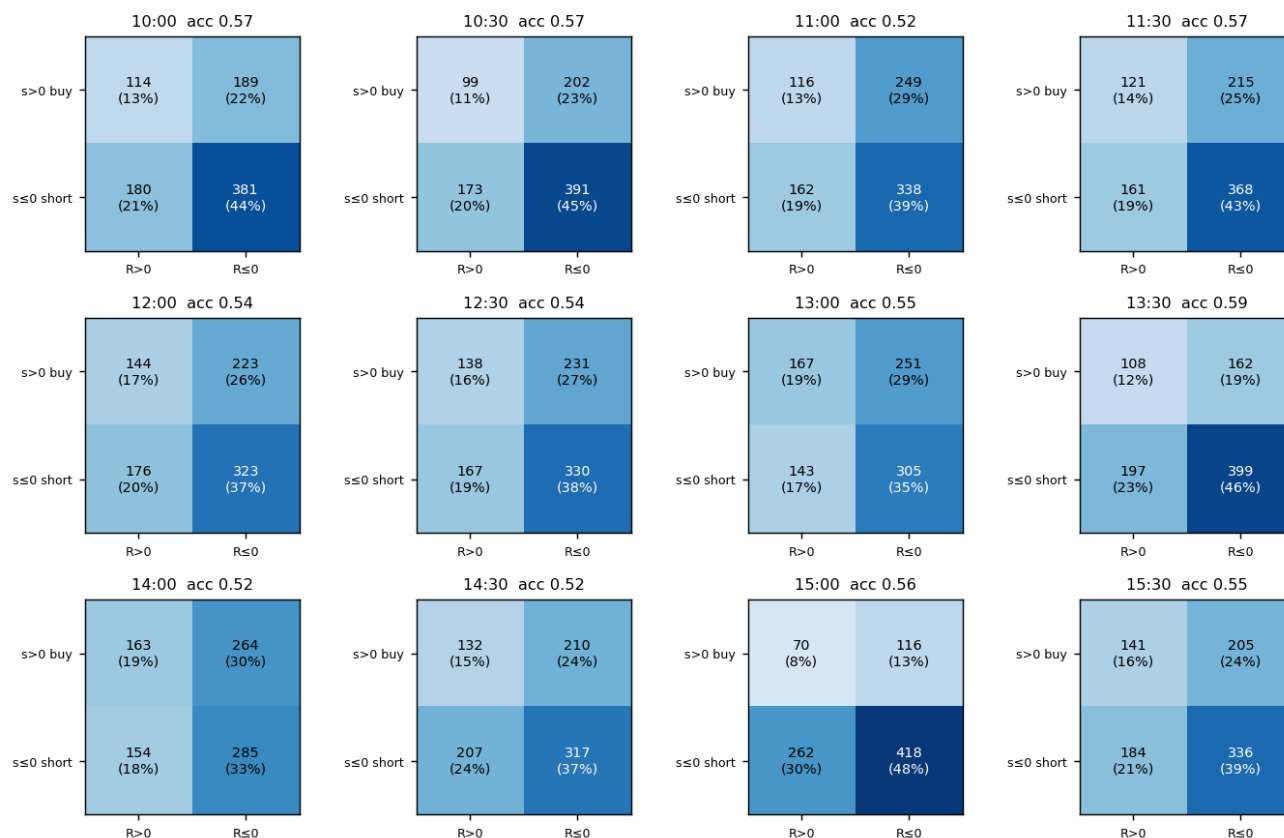


Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and $\text{Sharpe}_{\text{clk}}$ by entry clock (grouped bars). $\text{Sharpe}_{\text{clk}}$ is that clock's daily series $\times \sqrt{252}$, not the pooled $\text{Sharpe}_{\text{ann}}$. Delta-hedged $t \rightarrow T$: hold entry K to official close, rebalance Δ every 30 minutes.

sign(s) vs sign of long-package R (mid), block-diagonal ridge, 866 days



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-package midpoint return. 15:30 is the paper trade.

Each row is the daily sum of the twelve overlapping remaining-session books: enter at each clock 10:00–15:30 ET, hold those strikes to the official close, delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Notes

One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s twelve overlapping remaining-session books. That reserved name is the book’s daily P&L after pooling the session. The twelve books overlap; they are not twelve successive 30-minute holds. $\text{Sharpe}^\times$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^\times$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.